Time Series Foundation Model

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Background

Large Language Models (LLM)s have outperformed expert models in nearly all NLP tasks, from text classification to text generation. The ability to train such large models has been made possible by two equally crucial factors: (i) the abundance of available data, (ii) the parallelization of the training paradigm without sacrificing performance. While the first factor has long been available, the second factor was realized through the *attention mechanism* introduced in the transformer architecture.

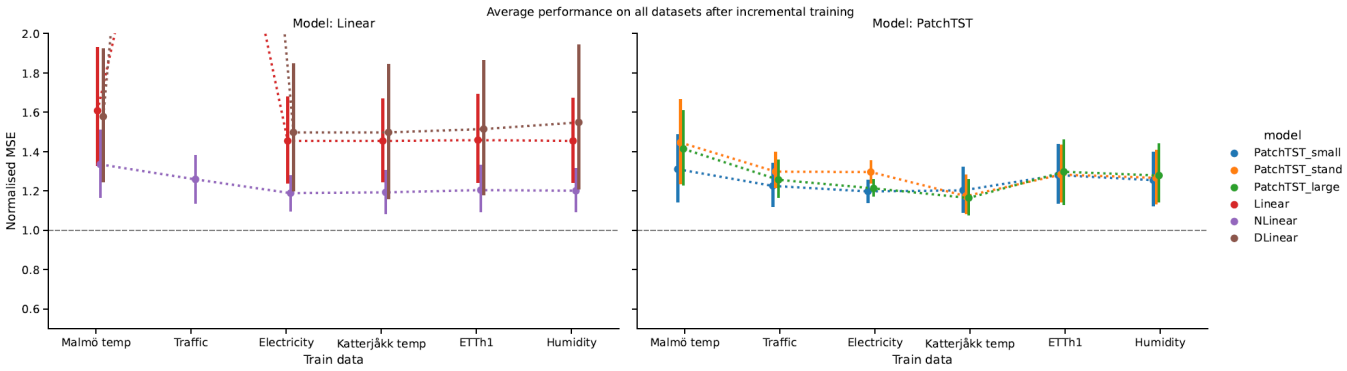
Given this context, it is natural to ask "Is it possible to apply the same technique to other types of sequential data, namely, time series?". Over the last few years, there has been numerous attempts to adapt similar architectures for time series forecasting. Researchers have achieved positive results with designs such as: **Informer**, **Autoformer**, **Pyraformer**, and **FEDformer**, among others. However in 2022, all these models were challenged by an *embarrassingly simple* **linear** model. The linear design not only outperformed the proposed transformer architecture but also was orders of magnitude faster to train on the same dataset.

To understand why similar approaches are not performing well for the time series forecasting, we need to grasp the concept of the unit of information, known as *token*. The attention mechanism is simple yet powerful, capable of finding connections between tokens within a context window. It is very natural to find a numerical representation for tokens to feed into a network. However, if one treats an individual record (a single number) in a time series as a token, not only does the training time scale with the length of context $\mathcal{O}(n^2)$, but also there is not much information in a single record to make a connection between them in the time series context. Therefore, the concept of *token* in time series must be introduced synthetically. In 2023, Nie et al. proposed a design called PatchTST, which naturally defines the idea of a token as a patch of a finite number of records. The attention mechanism can then work to find connections between these patches.

Relevant studies in our team

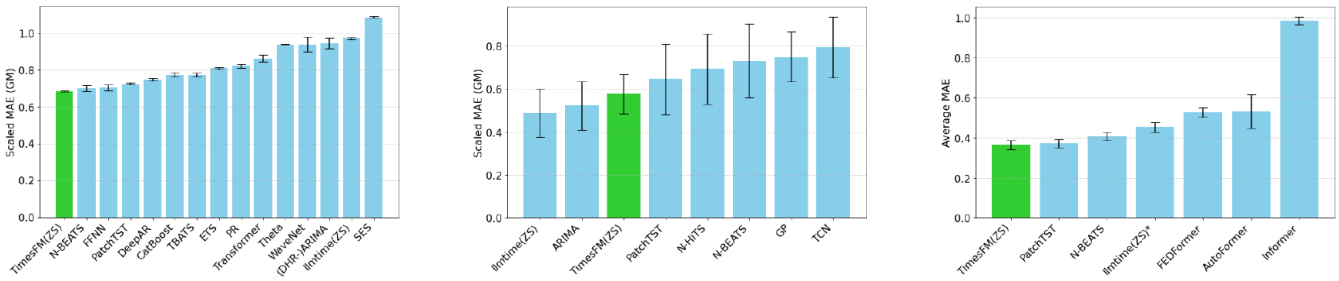
With positive results from PatchTST, a master student in our team conducted research to explore more relevant ideas:

- Benchmarking available architectures on datasets from various domains.
- Investigating PatchTST's potential to capture different components of time series.
- Assessing transferability of PatchTST from one domain to another.
- Evaluating PatchTST's architecture as a candidate for a time series foundation model.



The study also investigated the model's forecasting performance when applied to human-behavior-based time series (e.g., traffic) versus natural-process-based time series (e.g., weather factors).

During the research period, a new technique has been developed at Google, resulting in a foundation model called **TimesFM** that, according to their paper, can outperform the existing solutions including PatchTST

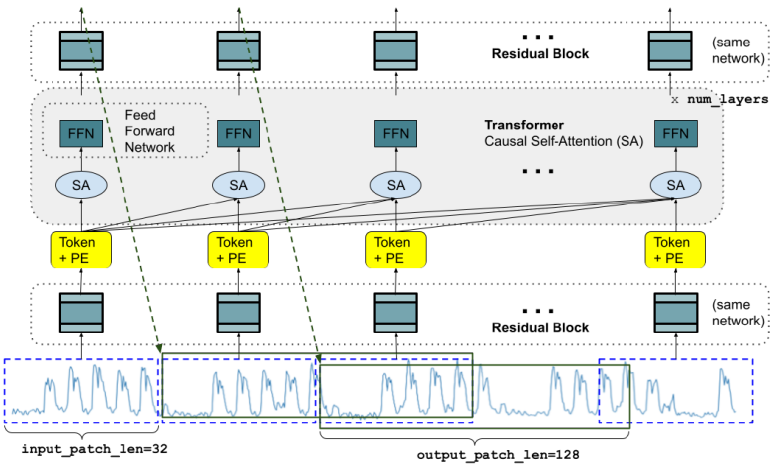


(a) Monash Archive [GBW+21] (b) Darts [HLP+22] (c) ETT (Horizons 96 and 192) [ZZP+21]

Research Questions

TimesFM is not the first attempt to train a time series foundation model (e.g., IBM, TimeGPT). However, since the model and the source code are open, it presents an opportunity to compare the architectures and investigate its performance against expert models. Additionally, like any other foundation model, it can be fine-tuned on a specific dataset, which is crucial for examining the model's transferability from one domain to another. Possible ideas that can be explored in this study include:

- **Architectural Comparison**: How different is the design of TimesFM versus PatchTST in terms of architecture? The patching idea seems an essential part of TimesFM model as well.



- **Off-the-shelf comparison**: How do the two models compare in terms of forecasting accuracy on different domain-specific time series data?
- **Fine-tunning**: If both approaches are comparable in performance, is there any advantage regarding fine-tunning time?
- **Training order**: What is the order of exposing domain-specific time series data during the training of TiemsFM? and, Are the models sensitive to the order?

Financial Domain Exploration

We would like to explore the capabilities of different approaches when it comes to the financial domain. A series of experiments can be suggested, such as:

- Comparing the performance of PatchTST and TimesFM expert models (i.e., when the architecture is trained on a specific dataset and tested on the same dataset).
- Applying PatchTST and TimesFM foundation models out-of-the-box for financial data forecasting.
- Fine-tuning PatchTST and TimesFM foundation models fine-tuned on financial data and testing them on data from the same domain.

A bonus idea that might be out of the scope of this research proposal, is the possibility of adjusting forecasting according to external events. Certain sudden phenomena, whether natural or human-triggered (e.g., epidemic disease, war, climate change), can dramatically affect time-dependent phenomena. Such sudden deviations from historical trend makes forecasting inaccurate. The question is that is there any way to design the network so that such events can be ingested to adjust the prediction.

Data

The set of time series data set that has been generally accepted as canonical and all new models are tested on is open and accessible to all. However such data does not contain any sample from financial domain. Part of this research is also to find open data set in financial domain. Account balance, company sales that eventually will be reflected in their bank accounts.

Refrencesses

- https://research.google/blog/a-decoder-only-foundation-model-for-time-series-forecasting/
- The timeFM paper: A DECODER-ONLY FOUNDATION MODEL FOR TIME-SERIES Forecasting
- IBM foundation model *TinyTimeMixer* (TTM) https://developer.ibm.com/tutorials/awb-foundation-model-time-series-forecasting/
- Tiny Time Mixers (TTMs): Fast Pre-trained Models Enhanced Zero/Few-Shot Forecasting of Multivariate Time Series
- TimeGPT-1: https://docs.nixtla.io/
- Foundation Models for Time Series Analysis: A Tutorial and Survey
- On the feasibility of Transformer based foundation models for time series forecasting, Arvid Gramer, Master's thesis
- Informer, Autoformer, Pyrformer, FEDformer
- Are transformers effective for time series forecasting?
- Link to the git for dataset.